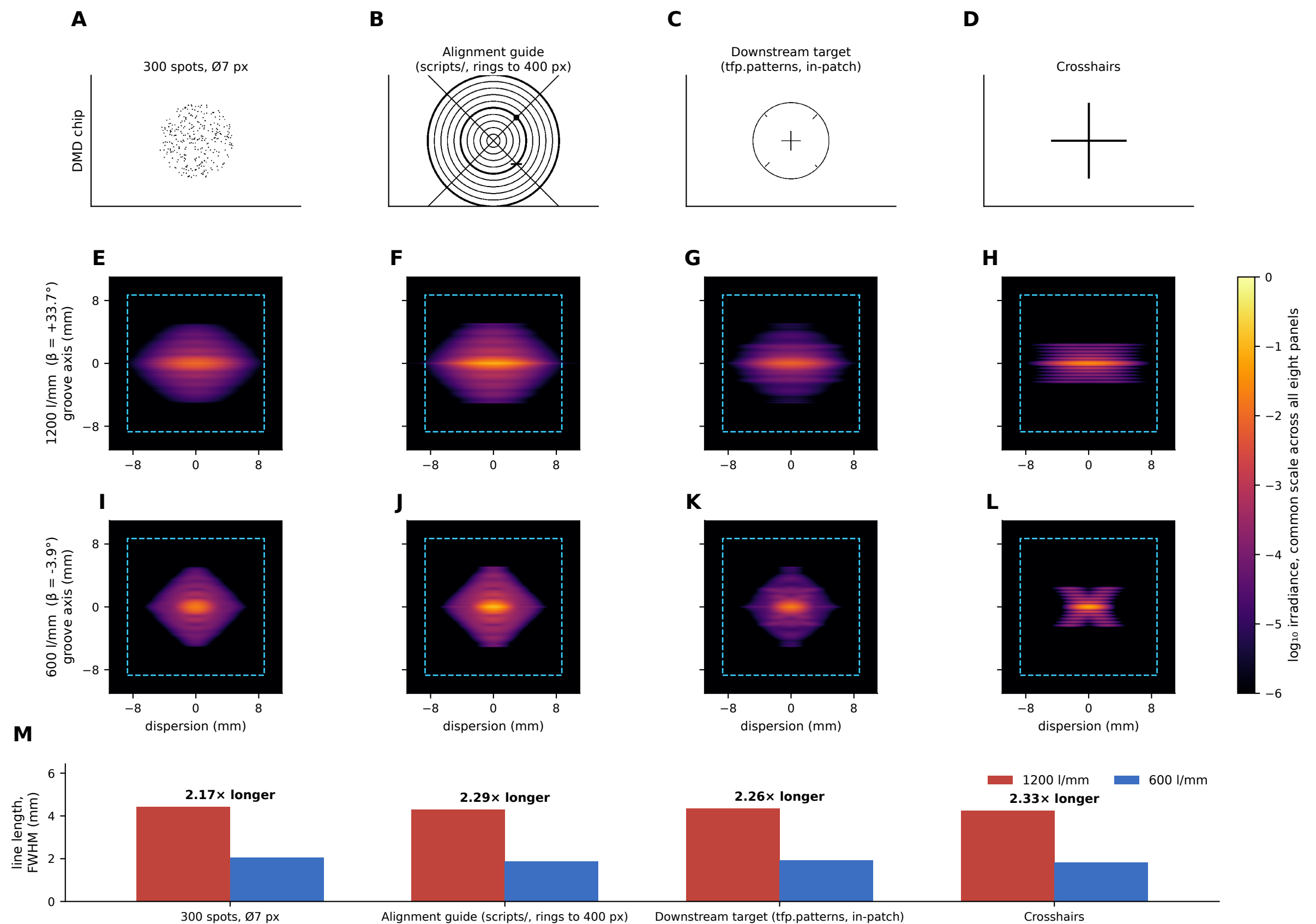




A 600 l/mm grating shortens the spectral line 2.2-2.3× — and brightens the liquid crystal by the same factor. Scalar Fourier propagation to the Meadowlark HSP1K at the bench-confirmed $f_7 = 300$ mm, with the grating as the only variable. β is solved from $\sin \alpha + \sin \beta = m\lambda/g$ at a fixed $\alpha = 43.7^\circ$ incidence, i.e. a straight swap into the same mount; the 1200 l/mm case returns $\beta = +33.7^\circ$, reproducing the handoff BOM, while 600 l/mm throws the order to $\beta = -3.9^\circ$, across the normal. **(A-D)** The frames written on the chip: 300 targets of Ø7 px (≈ 14 μ m at the sample); the bench alignment guide from scripts/alignmentTarget.m, whose rings deliberately run to 400 px radius, outside the Ø463 px patch, because locating the true illumination edge is its purpose; the patch-confined downstream target from tfp.patterns.alignmentTarget (edge ring, centre cross, long dispersion ticks, short groove ticks); and plain crosshairs. **(E-H)** Irradiance on the LC with the 1200 l/mm grating. **(I-L)** The same four frames with 600 l/mm. All eight share one log₁₀ colour scale normalised to the brightest panel and identical ± 11 mm axes, so the 600 l/mm row being visibly hotter is a real difference, not autoscaling; dashed cyan square is the 17.4 mm aperture. Chip-aligned features sit on the diagonals because the chip is clocked 45° and each transform is rotated into the lab frame. **(M)** Spectral-line length, measured as the FWHM of the dispersion-axis profile, paired per pattern. Mean shortening is 2.26×, against 2.35× higher peak irradiance — the same power in a shorter line. The underlying smear kernel shortens by 2.40×; measured lines fall below that because each pattern's own transform has finite width along dispersion, most visibly for the 300 spots (2.17×), whose broad envelope dominates the convolution. Because angular dispersion is $m \cdot g / \cos \beta$, halving the groove density gives 2.40× rather than 2×: $\cos \beta$ rises from 0.832 to 0.998 and returns some of what halving g took away. Caveat: re-angling the mount instead of swapping in place changes β and every ratio here.